MILITARY SPECIFICATION

ISOPROTERENOL SULFATE AEROSOL, NF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Isoproterenol Sulfate Aerosol, NF.

1.2 Classification. The aerosol covered by this specification shall be the following sizes (See 6.2).

Size 1	15 ml, 300 Metered Doses.
Size 2	22.5 ml, 450 Metered Doses.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard.
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.4).

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal, Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Isoproterenol Sulfate Aerosol in accordance with the tests standards and requirements of the NF including any supplements or revisions thereto. The Isoproterenol Sulfate Aerosol shall contain 2 mg of Isoproterenol Sulfate per ml of aerosol within the applicable assay limits. Each measured dose (adapter delivery) shall contain approximately 0.08 mg of Isoproterenol Sulfate within the applicable assay limits.

3.1.1 Measured doses. The Isoproterenol Sulfate Aerosol shall deliver a minimum of 300 measured doses for size 1 and a minimum of 450 measured doses for size 2 when determined as specified in 4.4.1.

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3.2 Ingredients. All ingredients entering into the finished inhalation shall be of USP or NF quality, or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.3 Expiration dating period. The expiration dating period shall be 48 months for size 1 and 60 months for size 2.

3.4 Delivery. Not less than 44 months for size 1 and 54 months for size 2 of the expiration dating period shall remain at time of delivery to the Government.

3.5 Workmanship. The material and its containers shall be free from any defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform and homogeneous quantity of Isoproterenol Sulfate Aerosol produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

Table I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of issue
Major	S-2	1.0	Package
Minor	II	2.5	Package

4.3.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101	Valve does not function properly (continuous or no spray)
102	Adapter clogged and/or does not spray.
103	Valve stem is loose in adapter.
104	Adapter not free of cracks or breaks.

Minor

201	Valve stem not straight.
202	Adapter not clean.
203	Adapter not free of physical defects affecting appearance.

* Inspection is not restricted to the classified possible defects listed above.

4.4 Tests.

4.4.1 Measured doses.

4.4.1.1 Size 1. Prepare the immediate container, with the adapter, as directed in the labeling. Atomize the inhalation as directed in the labeling and determine the weight of every tenth dose up to and including the 290th dose. Average the 29 dose weights. Determine the individual weights of the 291st through 300th dose. The average of the last 10 dose weights shall be not less than 97 percent of the average of the 29 dose weights and none of the last 10 dose weights shall be less than 90 percent of the average of the 29 dose weights.

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4.4.1.2 Size 2. Atomize the inhalation as directed in the labeling and determine the weight of every tenth dose up to and including the 440th dose. Average the 44 dose weights. Determine the individual weights of the 441st through 450th dose. The average of the last 10 dose weights shall be not less than 97 percent of the average of the 44 dose weights and none of the last 40 dose weights shall be less than 90 percent of the average of the 44 dose weights.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in Table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table II. Examination of Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged
Workmanship	Inadequate application of components such as incomplete closure of containers flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be Level A or C as specified (see 6.1).

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5.2.1 Level A.

5.2.1.1 Immediate container. The immediate container for size one or size two shall be a commercially available vial with a metering valve and rigid adaptor for inhalation therapy, considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.1.2 Unit package. Each unit consisting of an aerosal container and an oral adapter for the size specified shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.1.3 Intermediate package. Twelve unit packages of one size only shall be packed in an intermediate container constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequately secured to prevent accidental opening.

5.2.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (See 6.1).

5.3.1 Level A. Twenty four intermediate packages of one size only shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty four intermediate packages of one size only shall be packed in an exterior container as specified in paragraph 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

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5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit packages. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the National Stock Number.

NOTE: The National Stock Number shall not be preceded by any prefix.

Strip labeling may be used.

Label adhesion test shall not apply.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in paragraph 5.5.3 and 5.5.8.1 of PPP-C-186.

5.4.3 Exterior container. Exterior container shall be marked as specified in paragraphs 5.5.4 and 5.5.8.1 of PPP-C-186.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.3 and 5.4).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-014-8486	ISOPROTERENOL SULFATE AEROSOL, NF, 22.5 ml, 450 Metered Doses
6505-00-023-6481	ISOPROTERENOL SULFATE AEROSOL, NF, 15 ml, 300 Metered Doses

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6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

6.4 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1361
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-37211 ISOPROTERENOL SULFATE AEROSOL, NF

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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